COASTAL LANDFORMS / INDIA COAST AND CRZ

COASTAL-SYSTEM BALANCE → REFRACTION MAP → EROSIONAL SEQUENCE → CLIFF RETREAT SECTION → DEPOSITIONAL DISCRIMINATOR → RELATIVE SEA-LEVEL FORK

Read the thick cyan rail from Stage 00 to the final core synthesis. Each card uses a concept-appropriate structure; the grey E card is optional enrichment only and never repairs missing core content.

● PRIMARY CORE FLOW | ● SUBORDINATE EXTRA | ● APPROVAL / REVIEW

Approval: FALSE • Pending user review • active generation g2 • explicit approval of this exact generation is required • all prior artifacts unchanged

CORE BEFORE EXTRA • prior artefacts unchanged • exact same master drives poster and tiles

00

STAGE 00 COASTAL-SYSTEM BALANCE

COASTAL-SYSTEM BALANCE | WAVES + TIDES + CURRENTS | MARINE ENERGY | RIVERS + CLIFFS + OFFSHORE | SEDIMENT SUPPLY | SEA LEVEL + LAND MOTION | ACCOMMODATION SPACE | HUMAN WORKS

WAVES + TIDES + CURRENTS → marine energy → RIVERS + CLIFFS + OFFSHORE → sediment supply → SEA LEVEL + LAND MOTION → accommodation space → HUMAN WORKS → redistribute erosion and accretion

DEFINITION / COMPONENTS
- WAVES + TIDES + CURRENTS → marine energy
- RIVERS + CLIFFS + OFFSHORE → sediment supply

TRIGGER / MECHANISM
- SEA LEVEL + LAND MOTION → accommodation space
- HUMAN WORKS → redistribute erosion and accretion

EFFECT / INTERVENTION
- WAVES + TIDES + CURRENTS → marine energy
- RIVERS + CLIFFS + OFFSHORE → sediment supply

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: SEA LEVEL + LAND MOTION → accommodation space.

01

STAGE 01 REFRACTION MAP

REFRACTION MAP | DEEP WATER | WAVE FRONTS TRAVEL FASTER | SHALLOW HEADLAND | FRONTS SLOW AND CONVERGE | FRONTS DIVERGE AND ENERGY DISPERSES | EROSION FOCUS | DEPOSITIONAL SHELTER

DEEP WATER → wave fronts travel faster

SHALLOW HEADLAND → fronts slow and converge → BAY → fronts diverge and energy disperses → RESULT → erosion focus / depositional shelter

SYSTEM / DEFINITION
- DEEP WATER → wave fronts travel faster
- SHALLOW HEADLAND → fronts slow and converge

PROCESS / MECHANISM
- BAY → fronts diverge and energy disperses
- RESULT → erosion focus / depositional shelter

SPATIAL / INDIA PATTERN
- DEEP WATER → wave fronts travel faster
- SHALLOW HEADLAND → fronts slow and converge

HAZARD / LIMIT / RESPONSE
- BAY → fronts diverge and energy disperses
- RESULT → erosion focus / depositional shelter

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: BAY → fronts diverge and energy disperses; RESULT → erosion focus / depositional shelter.

02

STAGE 02 EROSIONAL SEQUENCE

EROSIONAL SEQUENCE | HYDRAULIC ACTION + ABRASION | WIDENING AND ROOF ATTACK | ROOF COLLAPSE

JOINT / FAULT → hydraulic action + abrasion → CAVE → widening and roof attack → ARCH → roof collapse → STACK → STACK → undercutting → STUMP

DEFINITION / COMPONENTS
- JOINT / FAULT → hydraulic action + abrasion
- CAVE → widening and roof attack

TRIGGER / MECHANISM
- ARCH → roof collapse → STACK
- STACK → undercutting → STUMP

EFFECT / INTERVENTION
- JOINT / FAULT → hydraulic action + abrasion
- CAVE → widening and roof attack

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: JOINT / FAULT → hydraulic action + abrasion.

03

STAGE 03 CLIFF RETREAT SECTION

CLIFF RETREAT SECTION | WAVE NOTCH | BASAL UNDERCUTTING | INSTABILITY AND COLLAPSE | DEBRIS REMOVAL | RENEWED ATTACK | WIDENING WAVE-CUT PLATFORM

DEFINITION / COMPONENTS
- WAVE NOTCH → basal undercutting
- OVERHANG → instability and collapse

TRIGGER / MECHANISM
- DEBRIS REMOVAL → renewed attack
- RETREAT → widening wave-cut platform

EFFECT / INTERVENTION
- WAVE NOTCH → basal undercutting
- OVERHANG → instability and collapse

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: DEBRIS REMOVAL → renewed attack; RETREAT → widening wave-cut platform.

• CAVE → widening and roof attack

• STACK → undercutting → STUMP

• CAVE → widening and roof attack

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: JOINT / FAULT → hydraulic action + abrasion.

03

STAGE 03

CLIFF RETREAT SECTION

CLIFF RETREAT SECTION · WAVE NOTCH · BASAL UNDERCUTTING · INSTABILITY AND COLLAPSE · DEBRIS REMOVAL · RENEWED ATTACK · WIDENING WAVE-CUT PLATFORM

DEFINITION / COMPONENTS

- WAVE NOTCH → basal undercutting
- OVERHANG → instability and collapse

TRIGGER / MECHANISM

- DEBRIS REMOVAL → renewed attack
- RETREAT → widening wave-cut platform

EFFECT / INTERVENTION

- WAVE NOTCH → basal undercutting
- OVERHANG → instability and collapse

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: DEBRIS REMOVAL → renewed attack; RETREAT → widening wave-cut platform.

04

STAGE 04

DEPOSITIONAL DISCRIMINATOR

DEPOSITIONAL DISCRIMINATOR · ONE END ATTACHED · CROSSES BAY OR MOUTH · WATER ENCLOSED BEHIND BAR · JOINS ISLAND TO LAND

DEFINITION / COMPONENTS	TRIGGER / MECHANISM	EFFECT / INTERVENTION
SPIT → one end attached	LAGOON → water enclosed behind bar	SPIT → one end attached
BAR → crosses bay or mouth	TOMBOLO → joins island to land	BAR → crosses bay or mouth

DEFINITION / COMPONENTS

- SPIT → one end attached
- BAR → crosses bay or mouth

TRIGGER / MECHANISM

- LAGOON → water enclosed behind bar
- TOMBOLO → joins island to land

EFFECT / INTERVENTION

- SPIT → one end attached
- BAR → crosses bay or mouth

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: LAGOON → water enclosed behind bar; TOMBOLO → joins island to land.

05

STAGE 05

RELATIVE SEA-LEVEL FORK

RELATIVE SEA-LEVEL FORK · LAND RISE · SEA FALL · RAISED BEACH + TERRACE · EXPOSED MARINE FORMS · LAND FALL · SEA RISE · DROWNED RIVER VALLEY

LAND RISE / SEA FALL → emergence

RAISED BEACH + TERRACE → exposed marine forms → LAND FALL / SEA RISE → submergence → RIA = drowned river valley / FJORD = glacial trough

DEFINITION / COMPONENTS

- LAND RISE / SEA FALL → emergence
- RAISED BEACH + TERRACE → exposed marine forms

TRIGGER / MECHANISM

- LAND FALL / SEA RISE → submergence
- RIA = drowned river valley / FJORD = glacial trough

EFFECT / INTERVENTION

- LAND RISE / SEA FALL → emergence
- RAISED BEACH + TERRACE → exposed marine forms

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RIA = drowned river valley / FJORD = glacial trough.

06

STAGE 06

STORM-SURGE CHAIN

STORM-SURGE CHAIN · CYCLONE WIND + LOW PRESSURE · WATER SETUP · TRACK + SHELF + BATHYMETRY · LOCAL AMPLIFICATION · HIGH TIDE + WAVES + RIVER FLOW · COMPOUND LEVEL · EXPOSURE + WEAK EVACUATION

CYCLONE WIND + LOW PRESSURE → water setup → TRACK + SHELF + BATHYMETRY → local amplification → HIGH TIDE + WAVES + RIVER FLOW → compound level → EXPOSURE + WEAK EVACUATION → disaster risk

DEFINITION / COMPONENTS

- CYCLONE WIND + LOW PRESSURE → water setup
- TRACK + SHELF + BATHYMETRY → local amplification

TRIGGER / MECHANISM

- HIGH TIDE + WAVES + RIVER FLOW → compound level
- EXPOSURE + WEAK EVACUATION → disaster risk

EFFECT / INTERVENTION

- CYCLONE WIND + LOW PRESSURE → water setup
- TRACK + SHELF + BATHYMETRY → local amplification

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: HIGH TIDE + WAVES + RIVER FLOW → compound level.

STAGE 06 STORM-SURGE CHAIN

STORM-SURGE CHAIN | CYCLONE WIND + LOW PRESSURE | WATER SETUP | TRACK + SHELF + BATHYMETRY | LOCAL AMPLIFICATION | HIGH TIDE + WAVES + RIVER FLOW | COMPOUND LEVEL | EXPOSURE + WEAK EVACUATION

CYCLONE WIND + LOW PRESSURE → water setup → TRACK + SHELF + BATHYMETRY → local amplification → HIGH TIDE + WAVES + RIVER FLOW → compound level → EXPOSURE + WEAK EVACUATION → disaster risk

DEFINITION / COMPONENTS

- CYCLONE WIND + LOW PRESSURE → water setup
- TRACK + SHELF + BATHYMETRY → local amplification

TRIGGER / MECHANISM

- HIGH TIDE + WAVES + RIVER FLOW → compound level
- EXPOSURE + WEAK EVACUATION → disaster risk

EFFECT / INTERVENTION

- CYCLONE WIND + LOW PRESSURE → water setup
- TRACK + SHELF + BATHYMETRY → local amplification

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: HIGH TIDE + WAVES + RIVER FLOW → compound level.

STAGE 07 INDIA COAST CONTRAST

INDIA COAST CONTRAST | BROADER PLAINS + LARGE DELTAS | NARROWER PLAIN + MANY ESTUARIES | PORT TEST | DEPTH, SHELTER, SILTATION, ENGINEERING | TENDENCY, NOT UNIVERSAL BINARY

SYSTEM / DEFINITION	PROCESS / MECHANISM	SPATIAL / INDIA PATTERN	HAZARD / LIMIT / RESPONSE
EAST → broader plains + large deltas	PORT TEST → depth, shelter, siltation, engineering	EAST → broader plains + large deltas	PORT TEST → depth, shelter, siltation, engineering
WEST → narrower plain + many estuaries	RULE → tendency, not universal binary	WEST → narrower plain + many estuaries	RULE → tendency, not universal binary

SYSTEM / DEFINITION

- EAST → broader plains + large deltas
- WEST → narrower plain + many estuaries

PROCESS / MECHANISM

- PORT TEST → depth, shelter, siltation, engineering
- RULE → tendency, not universal binary

SPATIAL / INDIA PATTERN

- EAST → broader plains + large deltas
- WEST → narrower plain + many estuaries

HAZARD / LIMIT / RESPONSE

- PORT TEST → depth, shelter, siltation, engineering
- RULE → tendency, not universal binary

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: PORT TEST → depth, shelter, siltation, engineering; RULE → tendency, not universal binary.

STAGE 08 NCCR EVIDENCE BOUNDARY

NCCR EVIDENCE BOUNDARY | 1990-2018 ASSESSMENT | ABOUT 33.6 PERCENT | ABOUT 26.9 PERCENT | ABOUT 39.6 PERCENT; NOT AN ANNUAL RATE

DEFINITION / COMPONENTS	TRIGGER / MECHANISM	EFFECT / INTERVENTION
PERIOD → 1990-2018 assessment	ACCRETING → about 26.9 percent	PERIOD → 1990-2018 assessment
ERODING → about 33.6 percent	STABLE → about 39.6 percent; not an annual rate	ERODING → about 33.6 percent

DEFINITION / COMPONENTS

- PERIOD → 1990-2018 assessment
- ERODING → about 33.6 percent

TRIGGER / MECHANISM

- ACCRETING → about 26.9 percent
- STABLE → about 39.6 percent; not an annual rate

EFFECT / INTERVENTION

- PERIOD → 1990-2018 assessment
- ERODING → about 33.6 percent

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: STABLE → about 39.6 percent; not an annual rate.

STAGE 09 CRZ IMPLEMENTATION LADDER

CRZ IMPLEMENTATION LADDER | CATEGORY-SPECIFIC NATIONAL RULES | HTL/LTL AND TECHNICAL INPUTS | CONSULTATION + APPROVAL + LOCAL MAP | APPRAISAL, DECISION, COMPLIANCE

NOTIFICATION → category-specific national rules

MAPPING → HTL/LTL and technical inputs → CZMP → consultation + approval + local map → PROJECT → appraisal, decision, compliance

DEFINITION / COMPONENTS

- NOTIFICATION → category-specific national rules
- MAPPING → HTL/LTL and technical inputs

TRIGGER / MECHANISM

- CZMP → consultation + approval + local map
- PROJECT → appraisal, decision, compliance

EFFECT / INTERVENTION

- NOTIFICATION → category-specific national rules
- MAPPING → HTL/LTL and technical inputs

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: CZMP → consultation + approval + local map.

09

STAGE 09

CRZ IMPLEMENTATION LADDER

CRZ IMPLEMENTATION LADDER | CATEGORY-SPECIFIC NATIONAL RULES | HTL/LTL AND TECHNICAL INPUTS | CONSULTATION + APPROVAL + LOCAL MAP | APPRAISAL, DECISION, COMPLIANCE

NOTIFICATION → category-specific national rules

MAPPING → HTL/LTL and technical inputs → CZMP → consultation + approval + local map → PROJECT → appraisal, decision, compliance

DEFINITION / COMPONENTS
- NOTIFICATION → category-specific national rules
- MAPPING → HTL/LTL and technical inputs

TRIGGER / MECHANISM
- CZMP → consultation + approval + local map
- PROJECT → appraisal, decision, compliance

EFFECT / INTERVENTION
- NOTIFICATION → category-specific national rules
- MAPPING → HTL/LTL and technical inputs

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: CZMP → consultation + approval + local map.

10

STAGE 10

COASTAL RESPONSE HIERARCHY

COASTAL RESPONSE HIERARCHY | KEEP NEW EXPOSURE OUT OF HIGH-RISK SPACE | BUFFERS + RESILIENT USE | MOVE ASSETS WHERE PERSISTENCE FAILS | SELECTIVE WORKS WITH DOWNDRIFT AUDIT

AVOID → keep new exposure out of high-risk space

RESTORE / ACCOMMODATE → buffers + resilient use → RETREAT → move assets where persistence fails → PROTECT → selective works with downdrift audit

DEFINITION / COMPONENTS
- AVOID → keep new exposure out of high-risk space
- RESTORE / ACCOMMODATE → buffers + resilient use

TRIGGER / MECHANISM
- RETREAT → move assets where persistence fails
- PROTECT → selective works with downdrift audit

EFFECT / INTERVENTION
- AVOID → keep new exposure out of high-risk space
- RESTORE / ACCOMMODATE → buffers + resilient use

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RESTORE / ACCOMMODATE → buffers + resilient use.

11

STAGE 11

COAST ANSWER SPINE

COAST ANSWER SPINE | DEFINE + MAP | PROCESS AND COASTAL SECTOR | ENERGY, SEDIMENT AND SEA-LEVEL CHAIN | PERIOD, CATEGORY, PLAN AND STATUS | INTEGRATED, ADAPTIVE AND LOCALLY MAPPED

DEFINE + MAP → process and coastal sector → DERIVE → energy, sediment and sea-level chain → VERIFY → period, category, plan and status → VERDICT → integrated, adaptive and locally mapped

DEFINITION / COMPONENTS
- DEFINE + MAP → process and coastal sector
- DERIVE → energy, sediment and sea-level chain

TRIGGER / MECHANISM
- VERIFY → period, category, plan and status
- VERDICT → integrated, adaptive and locally mapped

EFFECT / INTERVENTION
- DEFINE + MAP → process and coastal sector
- DERIVE → energy, sediment and sea-level chain

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: DEFINE + MAP → process and coastal sector.

E

EXTRA

SUBORDINATE ENRICHMENT — ONLY AFTER THE COMPLETE CORE

OPTIONAL ADVANCED ENRICHMENT | D.R. KHULLAR LISTS CYCLONE DANGERS AS GALES/STRONG WINDS, TORRENTIAL | KHULLAR NOTES MOST CASUALTIES FROM CYCLONES ARE CAUSED BY | TIDAL WAVES/STORM SURGES, WITH SEVERE SURGES PENETRATING INLAND FROM | LOW PRESSURE RAISES SEA LEVEL SLIGHTLY, BUT THE MAIN | IS STRONG ONSHORE CYCLONE WINDS PILING SEAWATER AGAINST A | INDIA LINK | THE SHALLOW, FUNNEL-SHAPED BAY OF BENGAL AND DELTAIC EAST

OPTIONAL PROCESS DEPTH
- D.R. Khullar lists cyclone dangers as gales/strong winds, torrential rain and high tidal waves, also called storm surges .
- Khullar notes most casualties from cyclones are caused by coastal inundation by
- tidal waves/storm surges, with severe surges penetrating inland from the coast.
- CAUTION: Mechanism: low pressure raises sea level slightly, but the main driver

DATA / INSTITUTION LIMIT
- is strong onshore cyclone winds piling seawater against a shallow coast.
- CAUTION: India link: the shallow, funnel-shaped Bay of Bengal and deltaic east coast make Odisha-West Bengal-Bangladesh especially surge-prone.
- Majid Husain describes tropical muddy coasts with mangroves in sediment-rich lagoons, bays, deltas and estuaries, and also mentions salt marshes.
- D.R. Khullar notes mangroves occur along India's sheltered estuaries, tidal creeks, backwaters, salt marshes and mudflats.

CONTEMPORARY PARALLEL
- CAUTION: Blue carbon = carbon stored by coastal and marine ecosystems,
- especially mangroves, seagrasses and salt marshes, in biomass and waterlogged sediments.
- CAUTION: India link: Sundarbans, Mahanadi-Godavari-Krishna deltas, Gulf of Kachchh, Gulf of
- Mannar and Andaman-Nicobar ecosystems link coastal protection with climate mitigation.

OPTIONAL STATUS — This optional depth is unnecessary for a competent core answer; use it only for added nuance after the complete core has been written.

END OF MASTER CANVAS • poster embeds this exact master • tiled pages are overlapping crops of the same pixels